

May 28, 2026 11:00 AM GMT

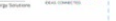
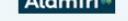
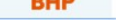
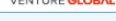
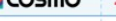
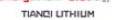
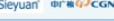
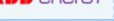
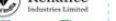
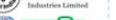
Investor Presentation | Asia Pacific

Energy Security & AI: Energy Meets Compute: Supercycle Recharges

Energy shocks have become more frequent, making energy and economic security critical in an AI world. US\$5trn+ investment needs should kick-start a golden age in dependable energy investments to secure AI, food, and tech supply chains after a decade of underinvestment. An investment supercycle unlocking US\$9trn in value beckons.

See our note [Asia Energy Security and AI: Energy Meets Compute: Supercycle Recharges](#)

Exhibit 1: Our 7 Key Surprises & Stocks Exposure Guide

Key Surprises	USA	CHINA	EMEA	INDIA	JAPAN	SOUTH KOREA	TAIWAN	ASEAN	AUSTRALIA
Surprise #1: Asia's energy capex will double by 2030.	   	  	   	  					
Surprise #2: The Return of fossil to Secure AI energy needs	 	 	  	 	 			 	  
Surprise #3: Diversification of energy sourcing to the US, LatAm	    		 		   	    			
Surprise #4: Domestic O&G production & refining picks up				  	  	 		   	   
Surprise #5: Energy storage expands rapidly		    				 			 
Surprise #6: Accelerated nuclear and power grid deployment	 	      	      	      		  			
Surprise #7: Corporate balance sheets can support 75% of investment				  	  			  	 

Source: Morgan Stanley Research

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Bottom Line: Best Energy Security and Powering AI Ideas

1. **Own dependable energy assets that are helping power AI:** Assets that give nations the ability to further energy self-sufficiency, such as nuclear, renewables, energy storage, greater domestic oil, and natural gas production and fuel refining.
2. **Diversification of energy sourcing to the US, LatAm, and within Asia:** Own tankers, LNG export and import infrastructure, and midstream pipeline assets.
3. **Accelerating nuclear power, fertilizer and power grid deployment,** as combating power shortages becomes even more critical. Data centers cross-subsidise households, with tiered power prices reducing affordability concerns.
4. **Oil and gas exploration picking up steam; coal gasification capacity builds set to double with return of coal-based power:** Own oil services and coal equipment manufacturers.
5. **Demand for metals and uranium likely to surprise:** Copper, aluminum, uranium, and nickel should see significant demand tailwinds from power and energy security infrastructure build.

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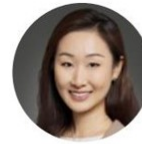
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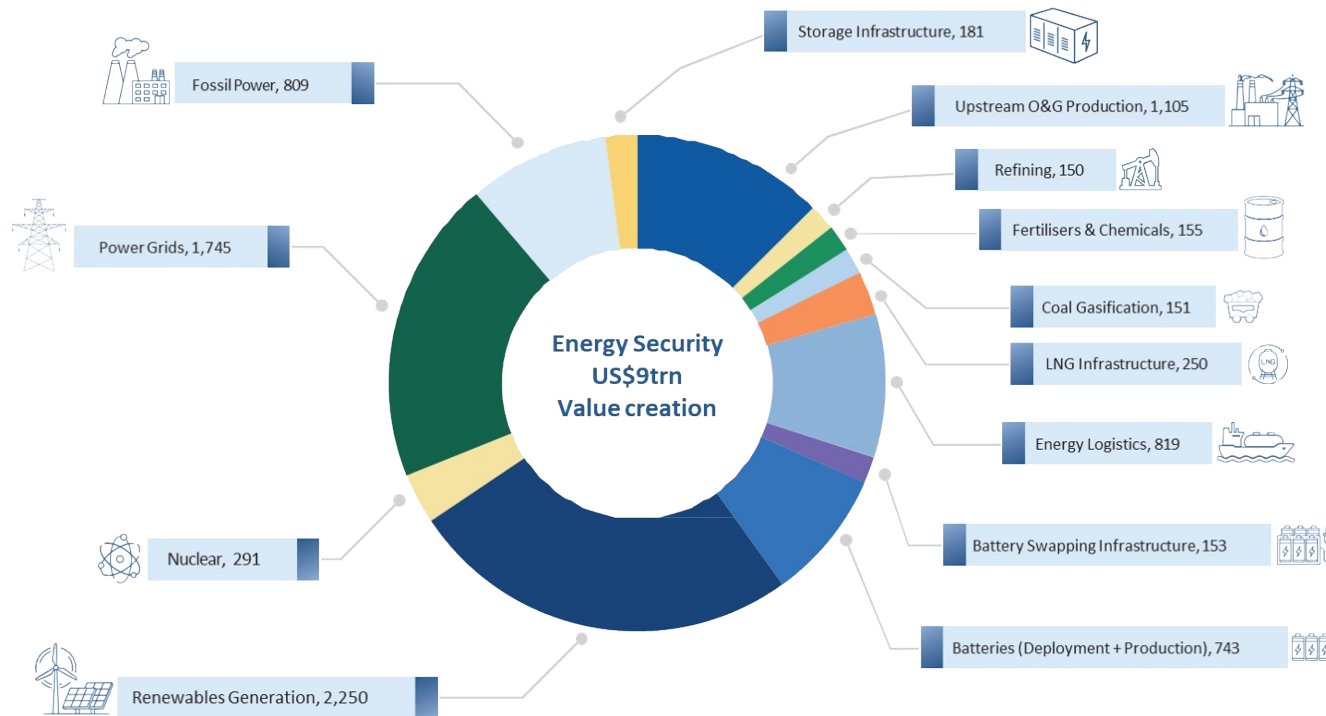
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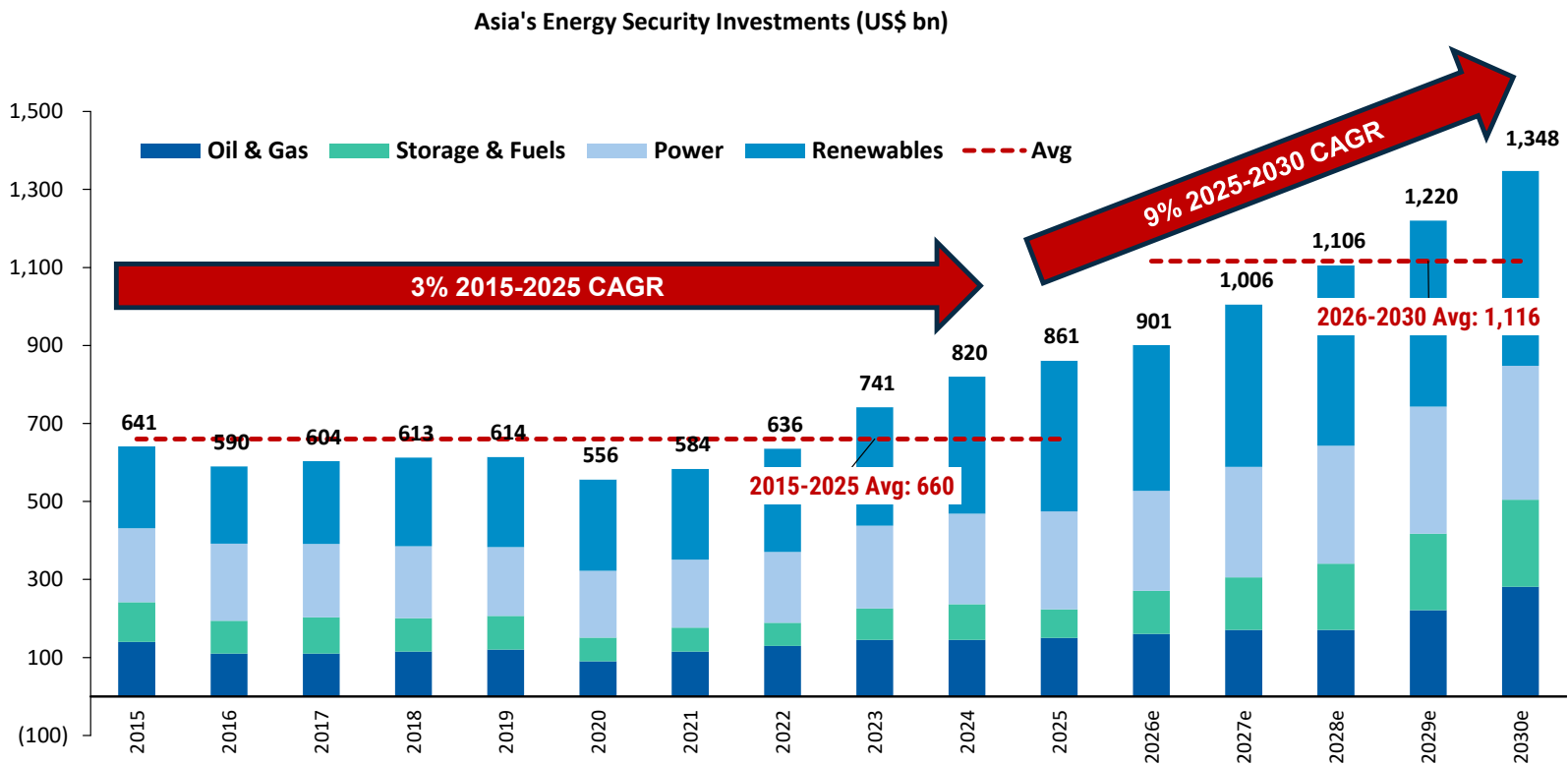
Energy Security & AI: The US\$9trn Value Creation Opportunity in Asia



We see a nearly 1.6x torque to the value creation as US\$5trn+ in investments across the region ensure security of tech, AI, food, and energy supply chains, unlocking US\$9trn in value.

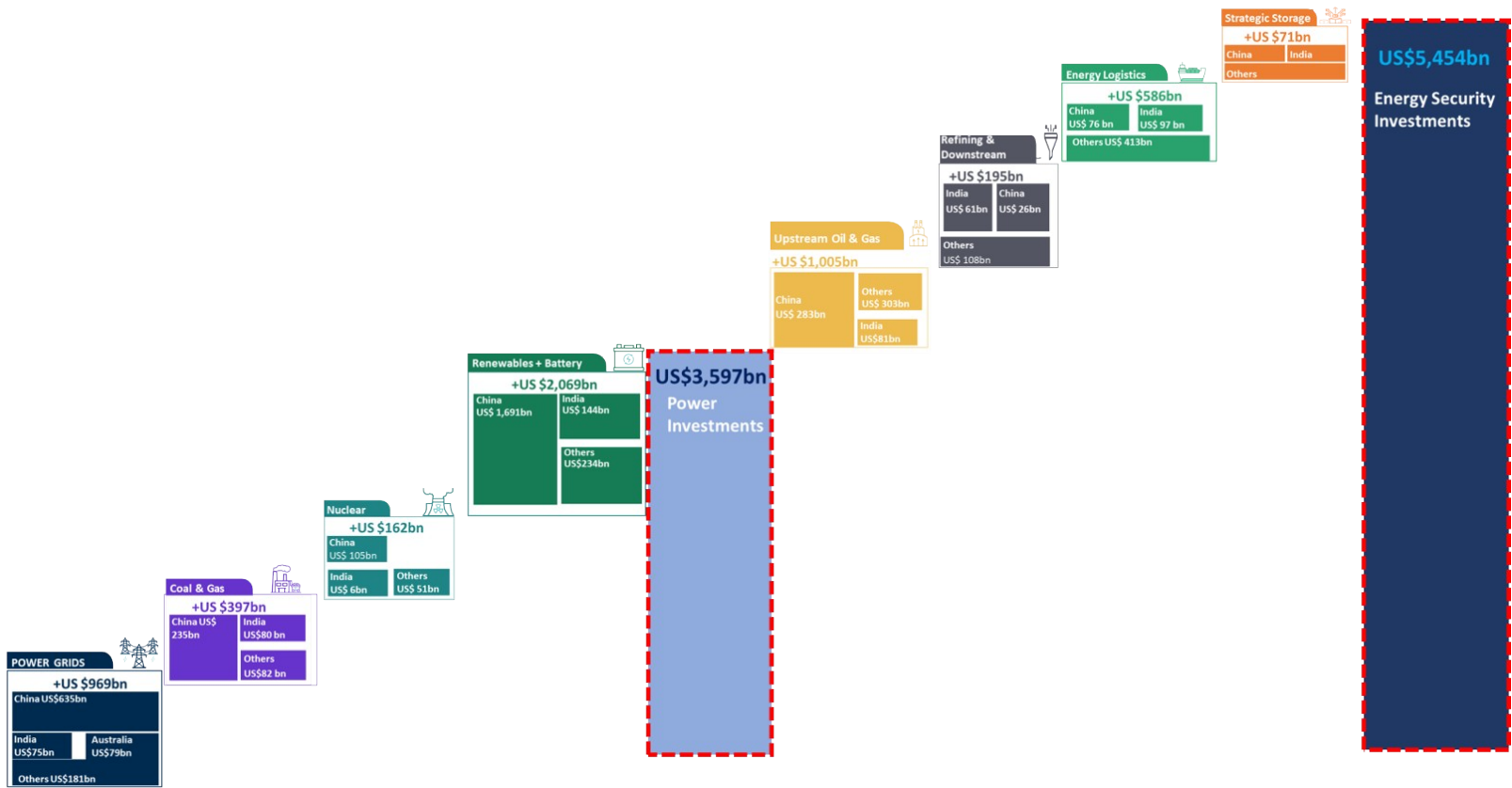
Source: Morgan Stanley Research estimates.

Surprise #1: Asia's Average Annual Energy Investments to Double by 2030



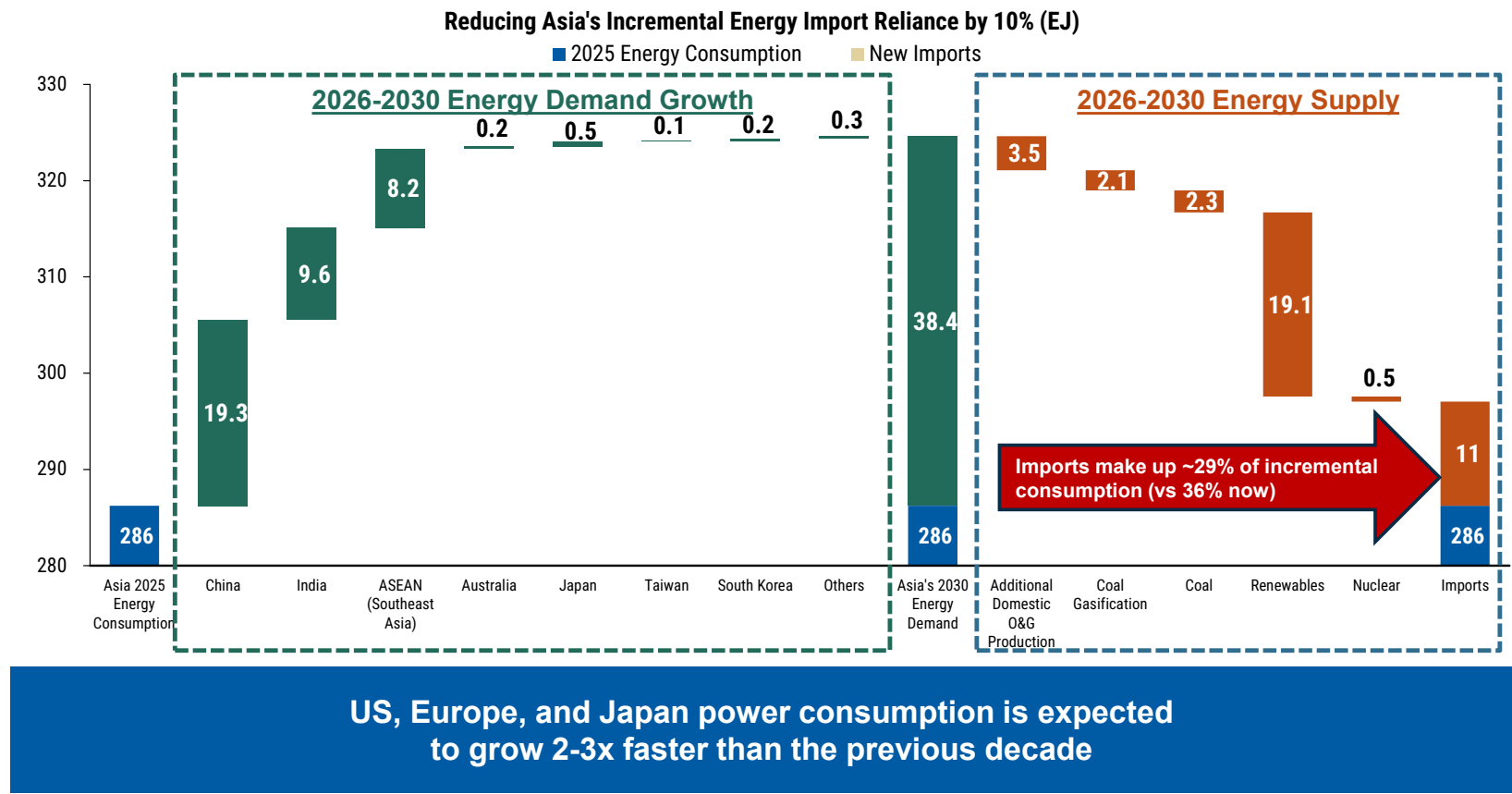
Source: Statistical Review of World Energy, Morgan Stanley Research estimates.

What's Driving the US\$5trn+ Supercycle?



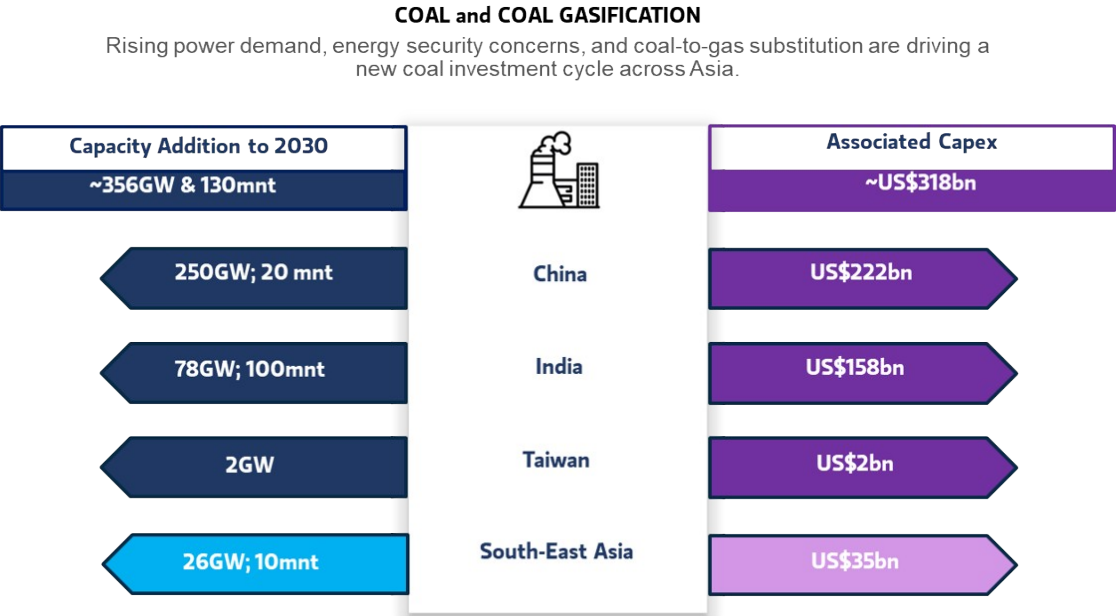
Source: Statistical Review of World Energy, Morgan Stanley Research estimates.

US\$1trn+ in New Investments = Lower Import Energy Reliance



Source: Statistical Review of World Energy, Morgan Stanley Research estimates.

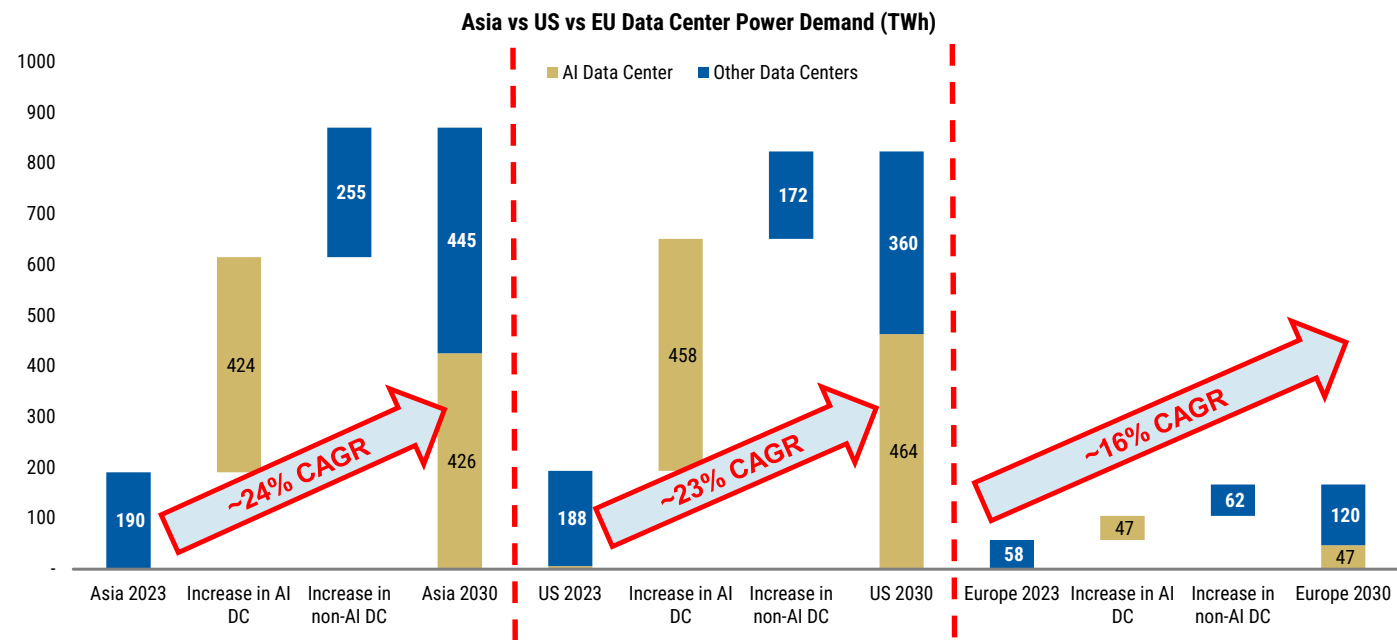
Surprise #2: The Return of Coal to Secure AI's Need for Energy



We expect Asian markets to spend US\$318bn in coal investments, driven by higher power demand and coal-to-gas switching

Source: Morgan Stanley Research estimates.

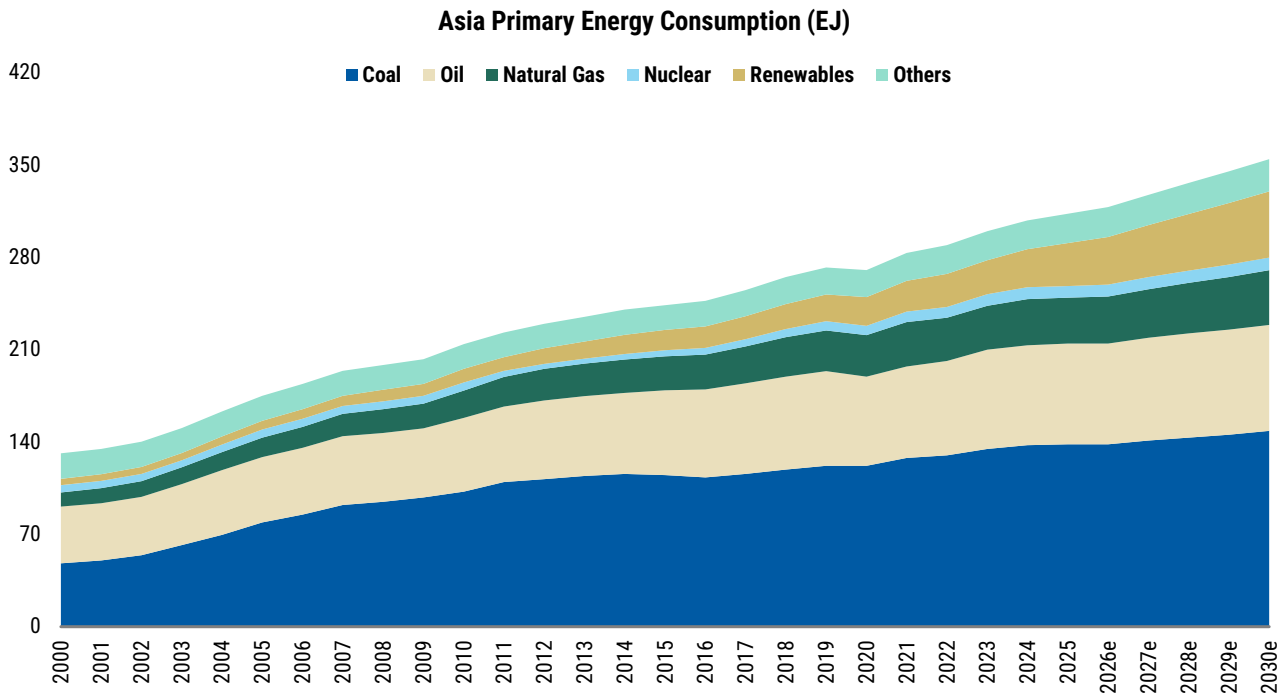
Powering AI: Asia vs US vs Europe Data Center Power Demand



We estimate Asia's DC power demand CAGR to be ~24% over 2023-30, exceeding that of the US and Europe, driven by faster AI inference demand and more investments committed to the region

Source: Morgan Stanley Research estimates.

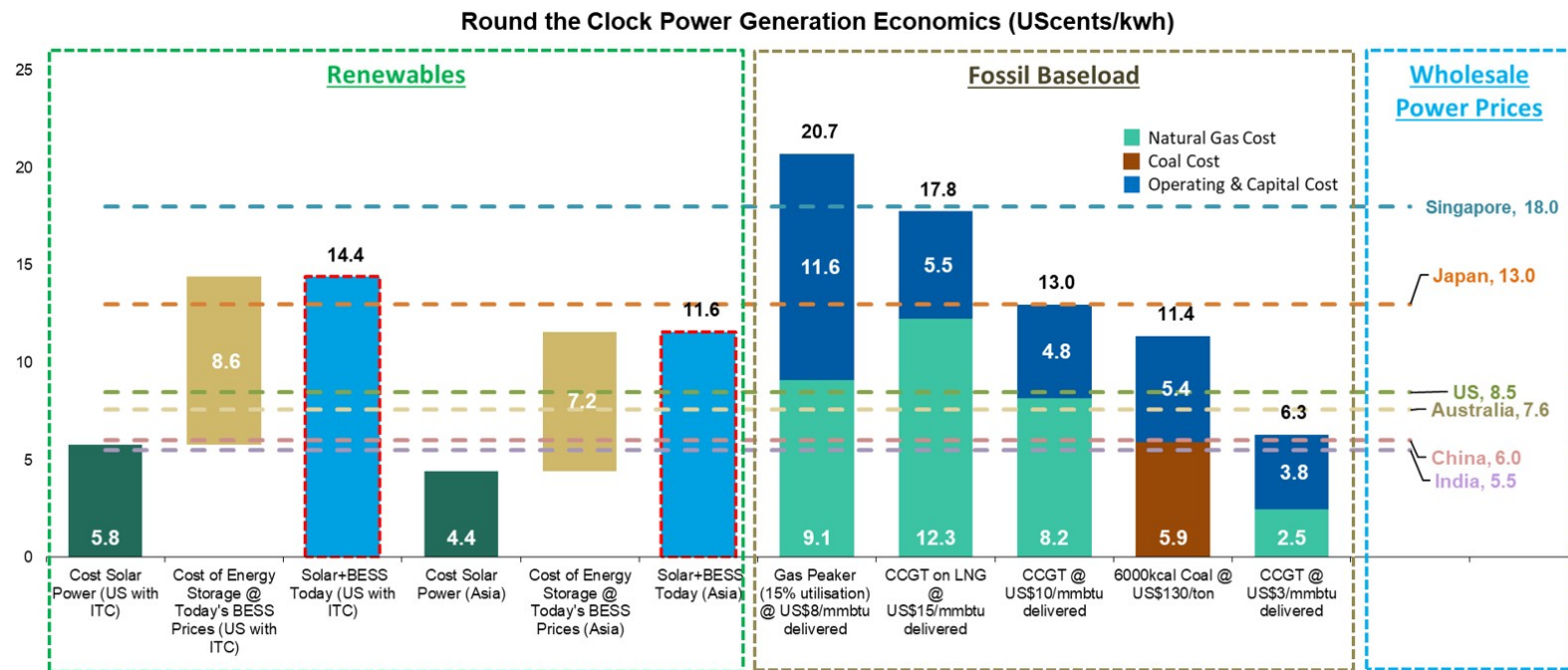
Asia’s Energy Consumption: Steady Growth with AI and Industrial Demand Tailwinds



Asia’s primary energy consumption to expand at a 2.7% CAGR through 2030, with powering AI at 5% of the energy mix.

Source: Statistical Review of World Energy, Morgan Stanley Research estimates.

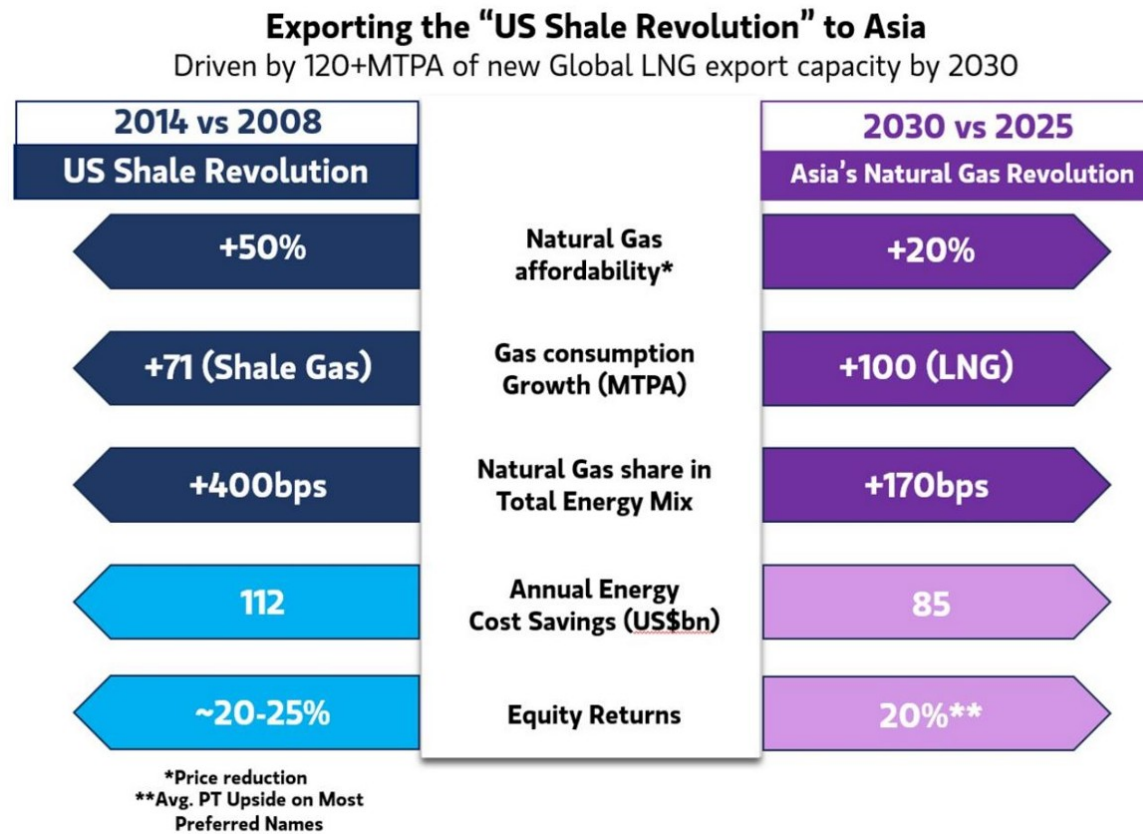
Comparing Power Generation Economics



US shale revolution will be exported to Asia for the first time at scale in 2027, with repercussions across the powering AI and energy security thematics

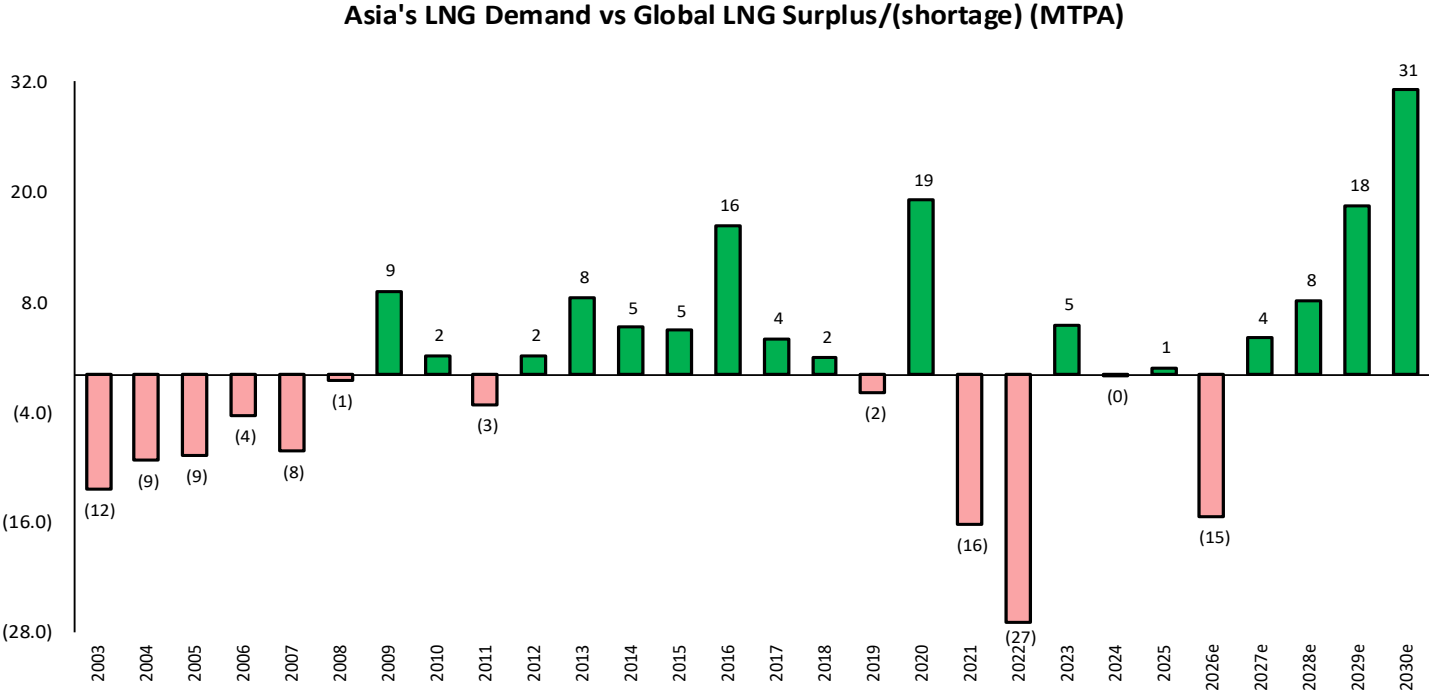
Source: Morgan Stanley Research estimates.

Surprise #3: Diversification of Energy Sourcing to the US & LatAm



Source: Morgan Stanley Research estimates.

The Supply Chain Shift: More US Energy for Power and Plastics



US shale revolution will be exported to Asia for the first time at scale in 2027, with repercussions across the powering AI and energy security thematics

Source: Statistical Review of World Energy, Morgan Stanley Research estimates.

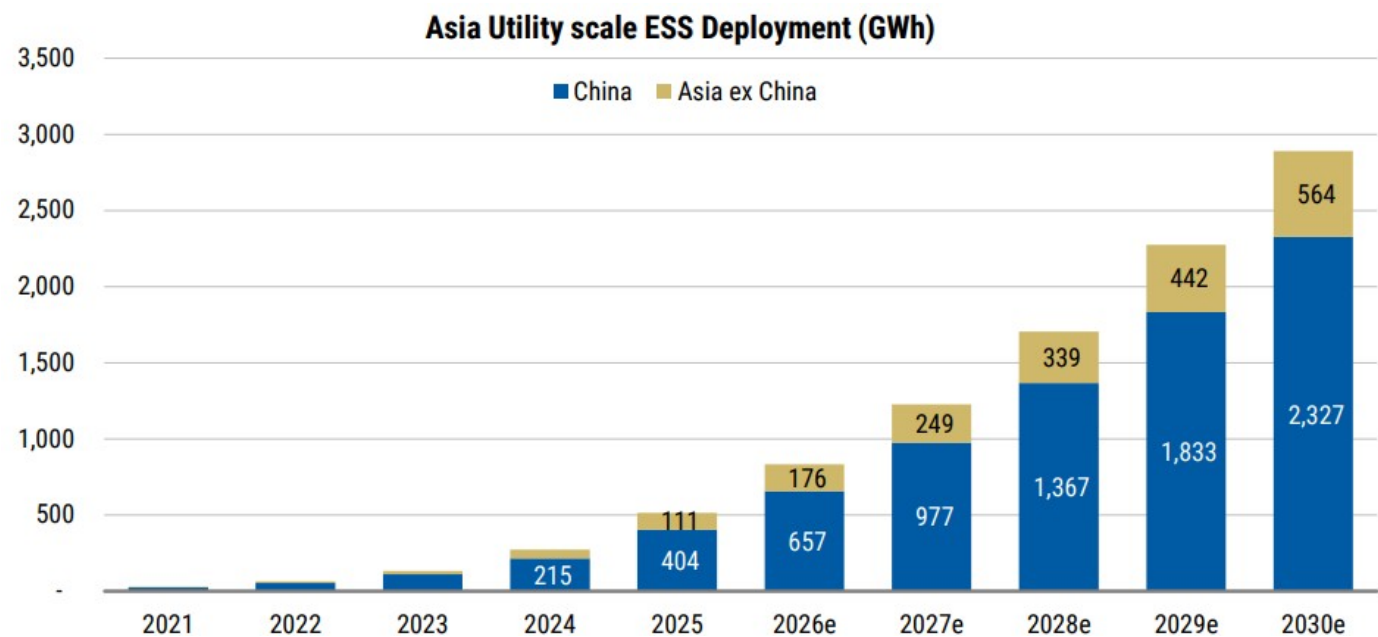
The Supply Chain Shift: Economies are Onshoring Critical Supply Chains

Capacity Company	F27						F28						Level of integration
	Polysilicon (Mtpa)	Ingot	Wafer	Solar cell	Module	Integrated	Polysilicon (Mtpa)	Ingot	Wafer	Solar cell	Module	Integrated	
Reliance	40,000			20.0	20.0	20.0	40,000			20.0	20.0	20.0	Polysilicon to module
Shri Sai Electricals		5	5	5	5			10	10	10	10		Polysilicon to module
Premier Energies			2	8	11	11		8	10	10	11	8	Ingot to module
Mundra Solar	30,000			10	10	10	30,000			10	10	10	Polysilicon to module
Waaree Energies		10	10	5	26	10		10	10	15	28	10	wafer/ingot to module
Renew Power				7	7	7				7	7	7	wafer to module
Tata Power Solar				4	4					4	4		Cell & Module
Renew-sys				3	6					3	6		Cell & Module
Vikram Solar				12	21				12	12	21	6 (F29)	Module
Saatvik Solar					4					5	9		
Goldi Solar				4	14					4	14		Cell & Module
Jupiter Solar				6	2					6	2		Cell & Module
Emmvee				3	7					3	7		Cell & Module
Websol				2	1					2	1		Cell & Module
Rayzon Solar					12						12		Module
First Solar					3						3		Module - vertically integrated
Others					20						40		
Total	70,000.0	15	17	87	171	58	70,000	28	42	108	203	55	

**India's solar manufacturing buildout is accelerating rapidly,
with major players building integrated facilities**

Source: Statistical Review of World Energy, Morgan Stanley Research estimates.

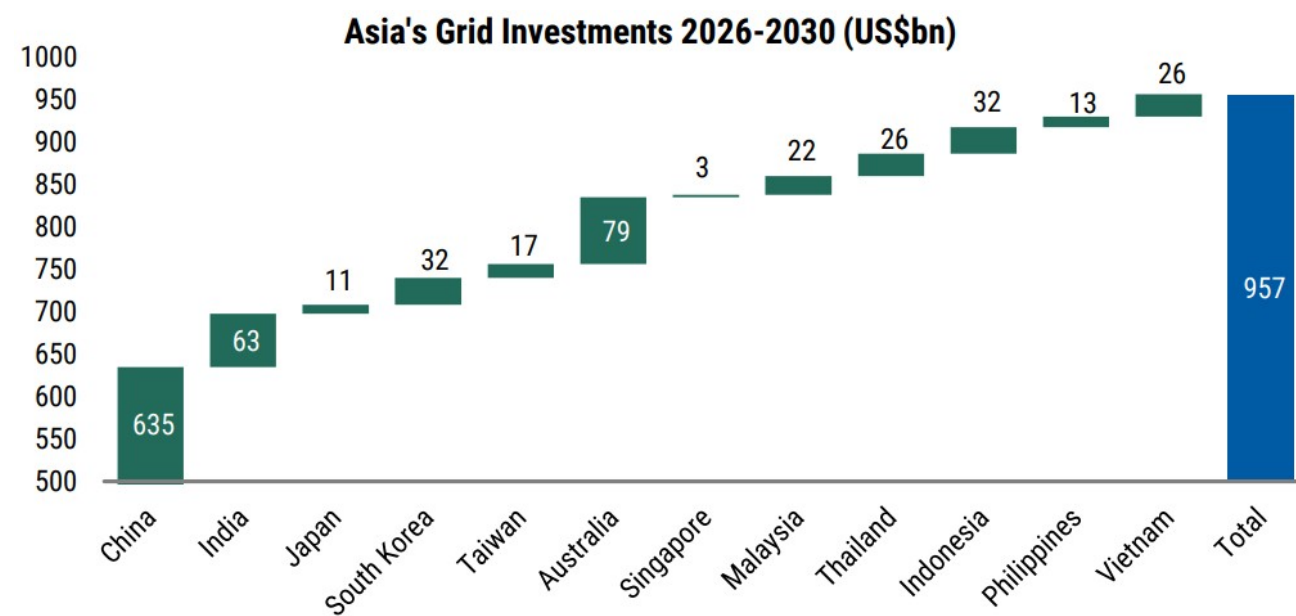
Surprise #4: Energy Storage Expands Rapidly, Even in Markets Outside of China



Coupled with high renewable deployment, we expect ESS additions of around 2700GWh by 2030

Source: Morgan Stanley Research estimates.

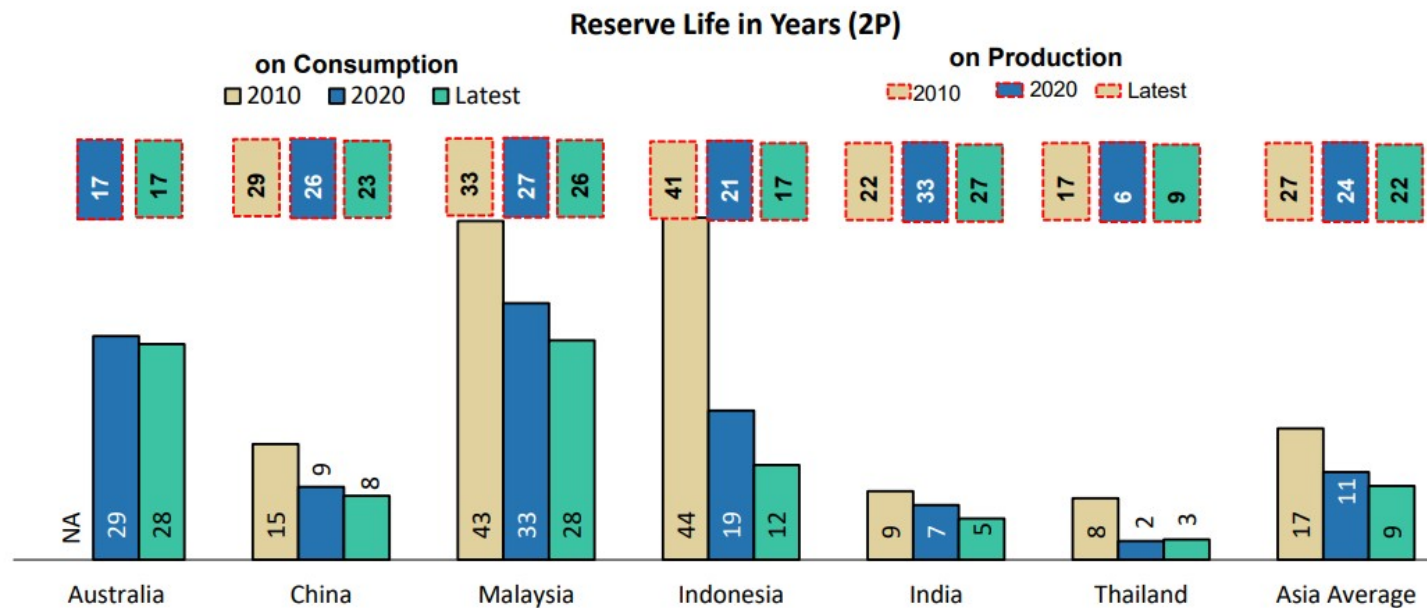
Surprise #5: Power Grid Deployment Boom



A lack of grid capacity is emerging as a critical bottleneck, driving higher levels of congestion and demand growth. Access to power via the grid is key in this 'Age of Electricity' and is a US\$1trn investment requirement.

Source: Morgan Stanley Research estimates.

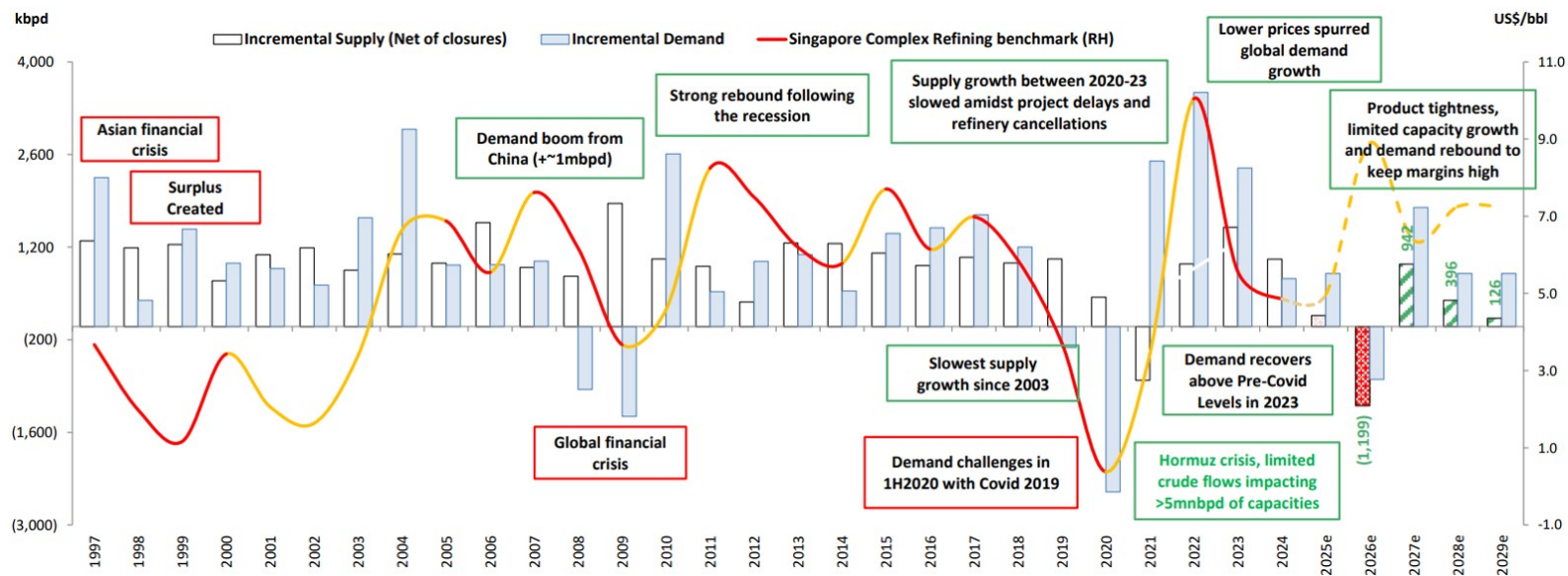
Surprise #6: Upstream Energy Picks up Steam & Coal Gasification Capacity Builds Double



Asia's reserve life based on energy consumption averages ~9-10 years (vs ~15 years on production), implying that Asia's reliance on external energy will remain elevated

Source: Morgan Stanley Research estimates.

Fuel: A New Refinery Needed Every Year



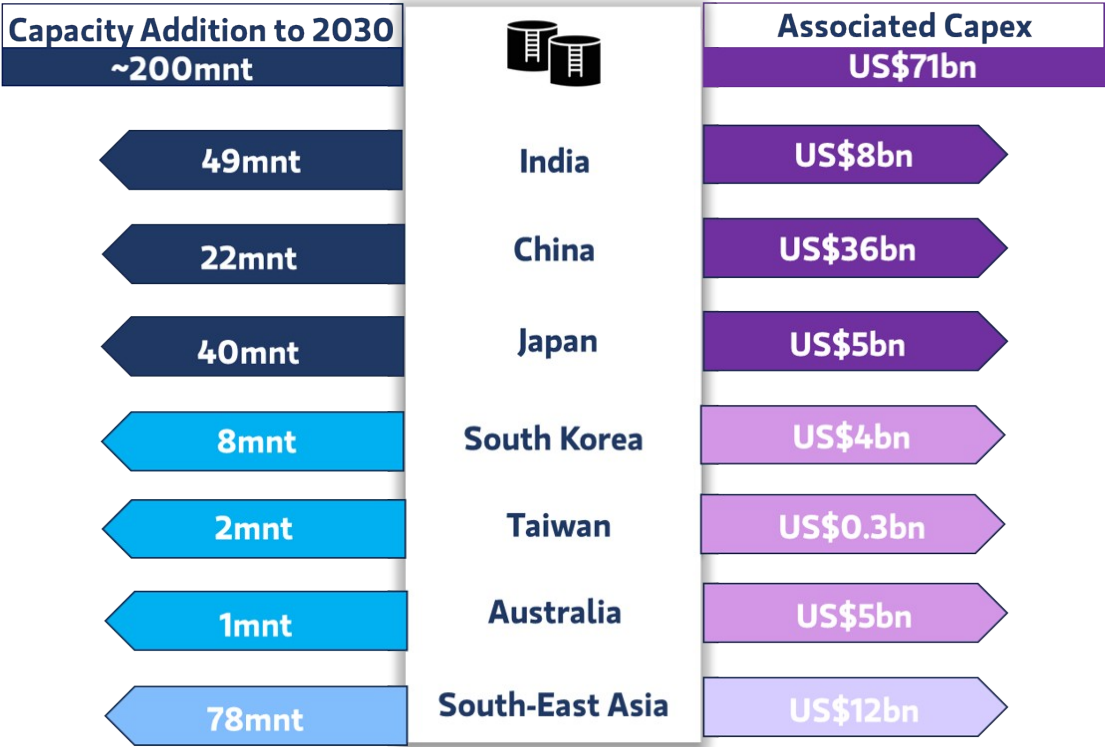
The refinery cycle in perspective: Much tighter as capacity growth remains below consumption growth by nearly 30-40%

Source: Statistical Review of World Energy, Morgan Stanley Research estimates.

Strategic Reserves: A Multi-year Capex Cycle

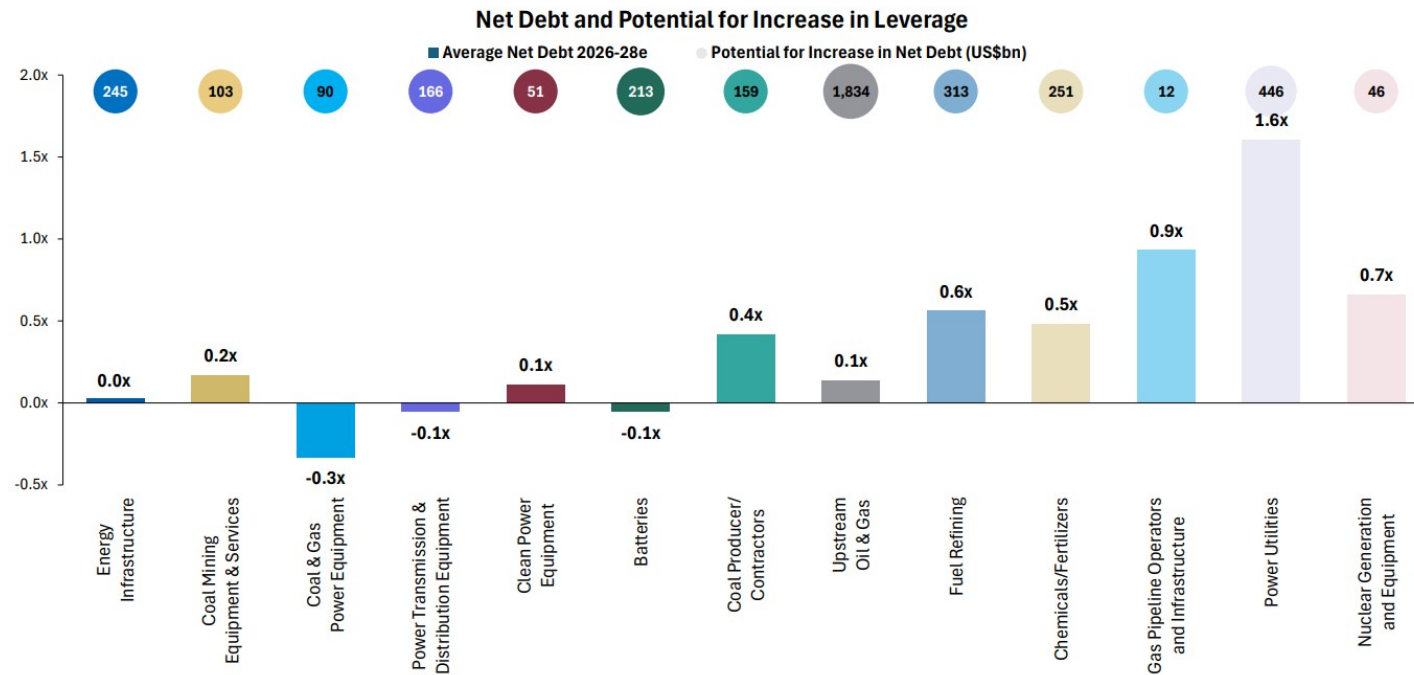
Storage Infrastructure

Rising energy security needs and the push to build strategic reserves are driving investments across Asia's storage infrastructure.



Source: Morgan Stanley Research estimates.

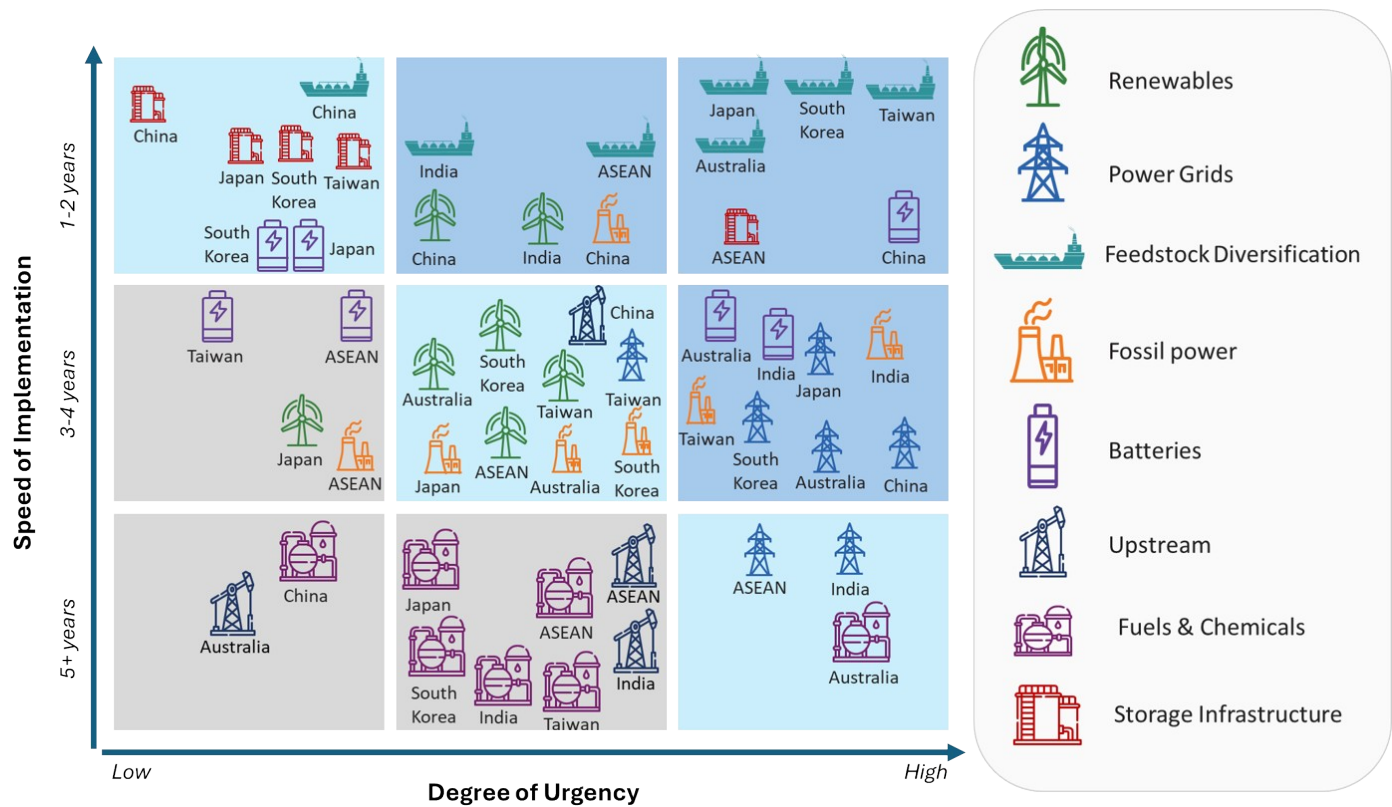
Surprise #7: Corporate Balance Sheets are Well Funded to Capitalize on Energy Security



Funding energy security: Corporate balance sheets are strong, with limited debt as companies have been behind the curve on growing investments vs consumption growth. Collectively, we see a US\$4trn potential for funding by debt on 2028 multiples

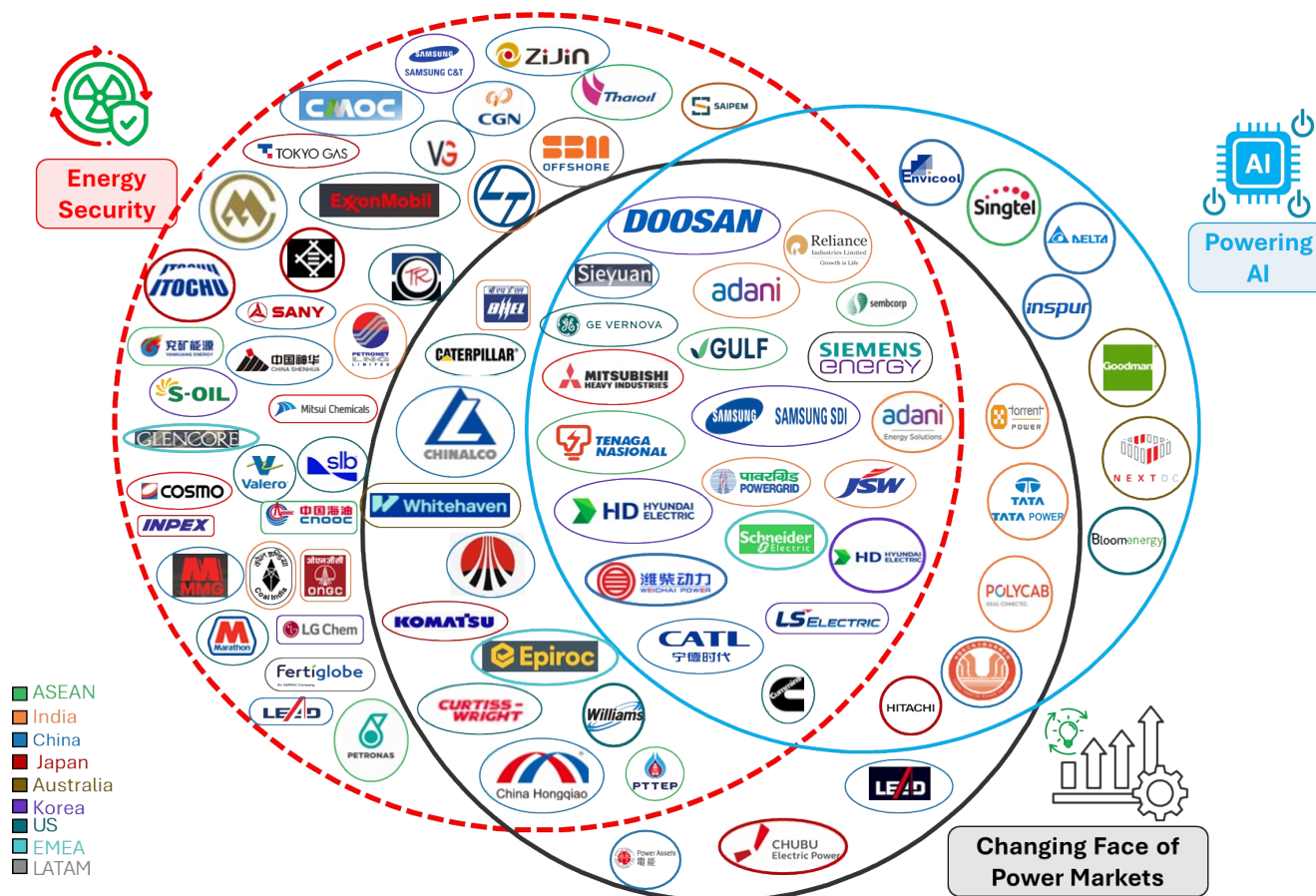
Source: Refinitiv, Morgan Stanley Research estimates.

Asia's Energy Security Implementation vs Urgency Matrix



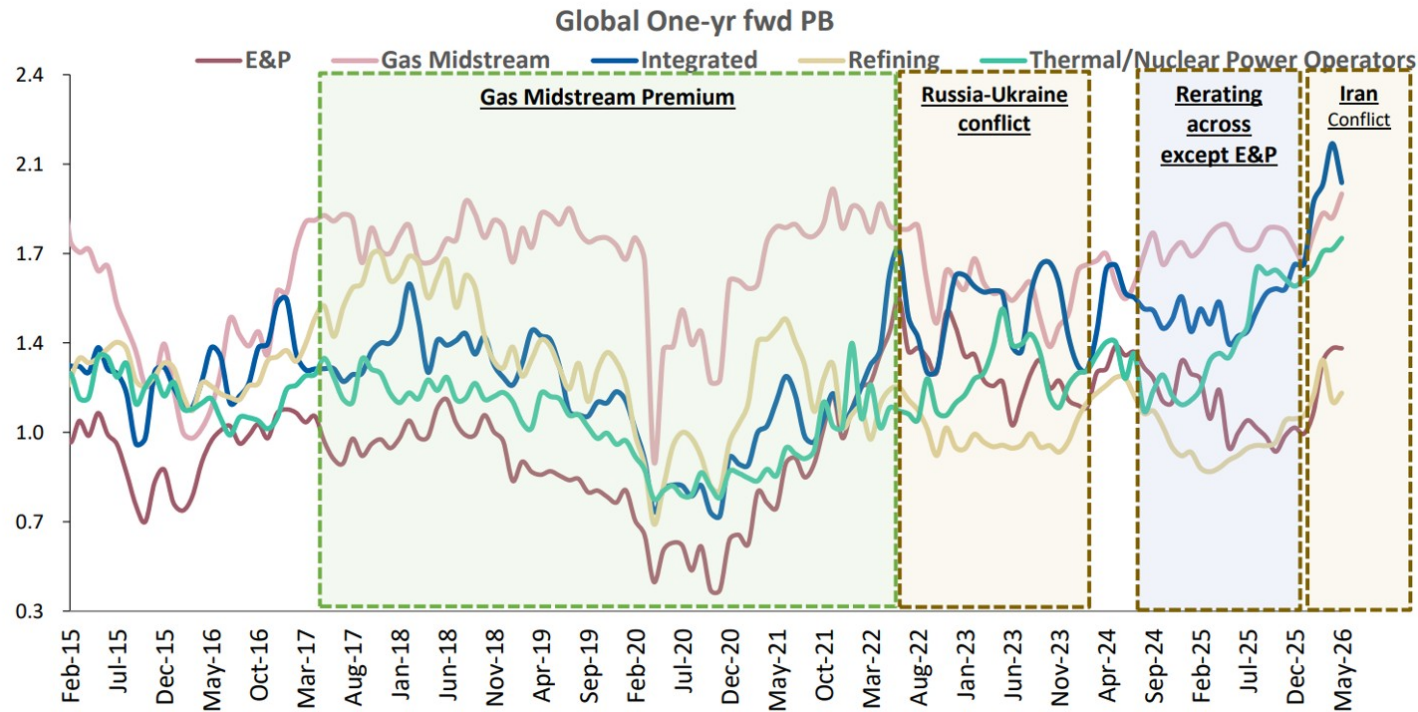
Source: Statistical Review of World Energy, Morgan Stanley Research estimates.

Buy “Dependable” Energy Supply Chain



Source: Morgan Stanley Research.

Learnings From Power Supply Chain Re-rating: Getting Broader Across Energy



Power supply chain has re-rated 20-50% in the past two years, as the consumption growth outlook has changed. The story widens to all dependable energy.

Source: Refinitiv, Morgan Stanley Research.

Asia Imports More Than a Third of its Energy Needs

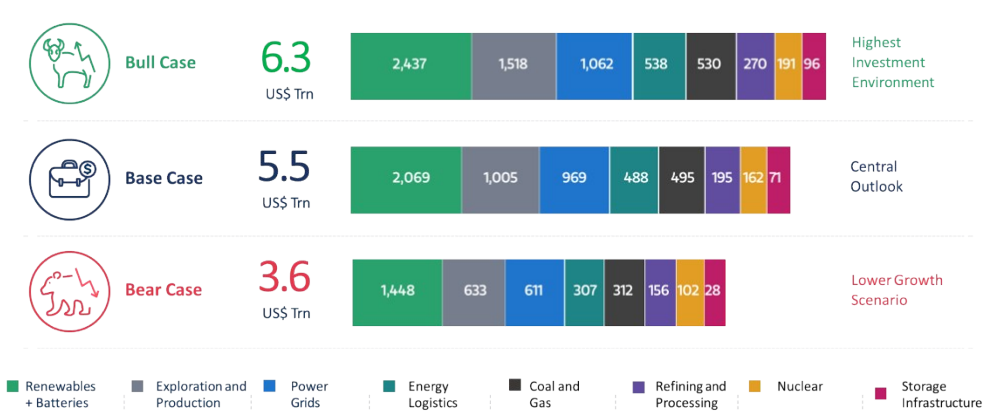
Power Generation Resiliency (% of Total Generation)	Imported				Self Reliance				
	Coal	Natural Gas	Nuclear	Total Imports	Coal	Natural Gas	Nuclear	Renewables & Others	Total Self Reliance
US	1%	2%	17%	20%	14%	38%	1%	27%	80%
Europe	7%	14%	19%	40%	2%	2%	2%	55%	60%
LATAM	1%	10%	NA	11%	1%	15%	2%	NA	18%
Middle East	0%	0%	NA	0%	0%	85%	5%	NA	90%
China	6%	1%	4%	11%	49%	2%	0%	38%	89%
India	3%	1%	3%	7%	66%	1%	0%	26%	93%
Japan	30%	34%	10%	74%	0%	0%	0%	26%	26%
Australia	0%	0%	NA	0%	44%	18%	0%	NA	61%
Singapore	NA	94%	NA	94%	NA	0%	0%	NA	0%
Malaysia	51%	12%	NA	63%	6%	28%	0%	NA	33%
Thailand	8%	21%	NA	29%	2%	38%	0%	NA	41%
Philippines	47%	8%	NA	54%	16%	9%	0%	NA	24%
South Korea	30%	29%	30%	88%	3%	2%	0%	7%	12%
Taiwan	29%	51%	1%	82%	0%	0%	0%	18%	18%
Vietnam	26%	7%	NA	33%	25%	3%	0%	NA	28%

Asia's needs for oil and natural gas for the transport and power sectors should see a shift toward coal, energy storage, and dependable fuels

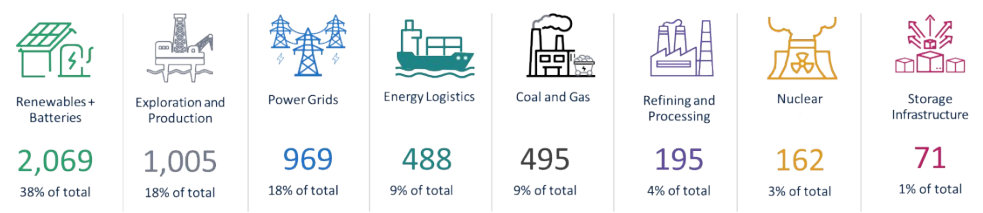
Source: Morgan Stanley Research estimates.

Energy Security: US\$5trn+ Investments Across Energy Verticals

Capital Investments (US\$ bn)



Investment Highlights (Base Case)



The US\$5trn+ investment supercycle: Dependable energy investments will nearly double vs the last decade, while renewables should plateau after five years of significant growth

Source: Statistical Review of World Energy, Morgan Stanley Research estimates.

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(as of April 30, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1546	42%	467	51%	30%	709	44%
Equal-weight/Hold	1568	43%	358	39%	23%	715	44%
Not-Rated/Hold	4	0%	0	0%	0%	1	0%
Underweight/Sell	555	15%	84	9%	15%	202	12%
Total	3,673		909			1627	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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INDUSTRY COVERAGE: India Oil & Gas

COMPANY (TICKER)	RATING (AS OF)	PRICE* (05/27/2026)
Mayank Maheshwari		
Bharat Petroleum Corp. (BPCL.NS)	O (10/02/2019)	Rs307.15
GAIL (India) (GAIL.NS)	E (03/23/2026)	Rs169.00
Gujarat Gas Ltd (GGAS.NS)	O (01/06/2023)	Rs369.80
Gujarat State Petronet Ltd (GSPT.NS)	U (09/12/2025)	Rs268.35
Hindustan Petroleum (HPCL.NS)	O (11/08/2023)	Rs402.90
Indian Oil Corp (IOC.NS)	O (12/10/2018)	Rs143.94
Indraprastha Gas Ltd. (IGAS.NS)	E (01/30/2025)	Rs164.61
Mahanagar Gas Ltd (MGAS.NS)	O (01/13/2025)	Rs1,103.20
Oil & Natural Gas Corp. (ONGC.NS)	O (09/20/2021)	Rs274.05
Oil India Limited (OILI.NS)	O (06/16/2022)	Rs488.00
Petronet LNG (PLNG.NS)	E (03/23/2026)	Rs278.80
Reliance Industries (RELI.NS)	O (07/12/2023)	Rs1,350.50
Vivek Rajamani		
Aarti Industries Limited (ARTI.NS)	U (09/03/2024)	Rs500.75
Deepak Nitrite Ltd (DPNT.NS)	O (05/03/2024)	Rs1,714.80
Navin Fluorine International Ltd. (NAFL.NS)	U (11/05/2025)	Rs7,244.50
PI Industries Ltd. (PIIL.NS)	O (06/16/2025)	Rs2,847.30
SRF Limited (SRFL.NS)	U (10/24/2024)	Rs2,736.80
Tata Chemicals Limited (TTCH.NS)	U (03/23/2026)	Rs774.55

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* Historical prices are not split adjusted.

INDUSTRY COVERAGE: ASEAN Utilities and Infrastructure

COMPANY (TICKER)	RATING (AS OF)	PRICE* (05/26/2026)
Mayank Maheshwari		
Airports of Thailand (AOT.BK)	O (08/25/2021)	Bt54.50
Global Power Synergy PCL (GPSC.BK)	U (09/12/2025)	Bt40.75
Gulf Development PCL (GULF.BK)	O (03/26/2025)	Bt62.50
Manila Electric Company (MER.PS)	O (06/20/2022)	PP600.00
Maynilad Water Services, Inc. (MYNLD.PS)	O (11/30/2025)	PP23.15
Perusahaan Gas Negara (PGAS.JK)	U (05/17/2023)	Rp1,895
SembCorp Industries Ltd (SCIL.SI)	O (03/23/2026)	S\$6.21
Tenaga Nasional (TENA.KL)	O (09/12/2023)	RM14.16

Vivek Rajamani

International Container Terminal Service (ICT.PS)

O (03/04/2024)

PP799.00

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INDUSTRY COVERAGE: ASEAN Energy and Materials

COMPANY (TICKER)	RATING (AS OF)	PRICE* (05/26/2026)
Mayank Maheshwari		
Alamtri Resources Indonesia (ADRO.JK)	U (01/18/2024)	Rp2,300
Bangchak Corporation Public Company (BCP.BK)	O (01/04/2026)	Bt34.50
Indorama Ventures PCL (IVL.BK)	O (08/09/2024)	Bt24.50
Keppel Ltd (KPLM.SI)	O (02/15/2024)	S\$10.84
Petronas Chemicals Group Berhad (PCGB.KL)	O (07/29/2025)	RM5.48
PTT Exploration & Production (PTTEP.BK)	O (02/11/2026)	Bt144.50
PTT Global Chemicals (PTTGC.BK)	O (07/29/2025)	Bt35.50
PTT Oil and Retail Business PCL (OR.BK)	E (08/09/2024)	Bt12.50
PTT Public Company (PTT.BK)	E (04/07/2026)	Bt36.50
Seatrium Limited (SEAT.SI)	U (01/16/2026)	S\$2.20
Siam Cement (SCC.BK)	O (09/06/2025)	Bt229.00
Star Petroleum Refining (SPRC.BK)	O (12/19/2018)	Bt7.35
Thai Oil Public Company (TOP.BK)	O (04/07/2026)	Bt46.50
United Tractors Tbk PT (UNTR.JK)	O (01/20/2026)	Rp23,800
Vivek Rajamani		
Indocement Tunggul Prakarsa Tbk PT (INTP.JK)	U (04/07/2025)	Rp4,990
Semen Indonesia (Persero) Tbk PT (SMGR.JK)	U (11/18/2024)	Rp1,775

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